

MATHEMATICAL INDUCTION

8.2 Principle of Mathematical Induction

The principle of mathematical induction is stated as follows:

If a proposition or statement $S(n)$ for each positive integer n is such that

- 1) $S(1)$ is true i.e., $S(n)$ is true for $n = 1$ and
- 2) $S(k + 1)$ is true whenever $S(k)$ is true for any positive integer k , then $S(n)$ is true for all positive integers.

Procedure:

1. Substituting $n = 1$, show that the statement is true for $n = 1$.
2. Assuming that the statement is true for any positive integer k , then show that it is true for the next higher integer.

For the second condition, one of the following two methods can be used:

- M_1 Starting with one side of $S(k + 1)$, its other side is derived by using $S(k)$.
- M_2 $S(k + 1)$ is established by performing algebraic operations on $S(k)$.

Example1: Use mathematical induction to prove that $3+6+9+\dots+3n=\frac{3n(n+1)}{2}$ for every positive integer n .

Solution: Let $S(n)$ be the given statement, that is,

$$S(n): 3+6+9+\dots+3n=\frac{3n(n+1)}{2}$$

1. When $n=1$, $S(1)$ becomes

$$S(1): 3 = \frac{3(1)(1+1)}{2} = 3$$

Thus $S(1)$ is true i.e., condition (1) is satisfied.

2. Let us assume that $S(n)$ is true for any $n=k \in N$, that is,

$$3+6+9+\dots+3k=\frac{3k(k+1)}{2}$$

The statement for $n=k+1$ becomes

$$\begin{aligned} 3+6+9+\dots+3k+3(k+1) &= \frac{3(k+1)[(k+1)+1]}{2} \\ &= \frac{3(k+1)(k+2)}{2} \end{aligned}$$

Adding $3(k+1)$ on both the sides of (A) gives

$$\begin{aligned} 3+6+9+\dots+3k+3(k+1) &= \frac{3k(k+1)}{2} + 3(k+1) \\ &= 3(k+1)\left(\frac{k}{2}+1\right) \\ &= \frac{3(k+1)(k+2)}{2} \end{aligned}$$

Example 2: Use mathematical induction to prove that for any positive integer n ,

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Solution: Let $S(n)$ be the given statement,

$$S(n): 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

1. If $n = 1$, $S(1)$ becomes

$$S(1): (1)^2 = \frac{1(1+1)(2 \times 1 + 1)}{6} = \frac{1 \times 2 \times 3}{6} = 1, \text{ which is true.}$$

Thus $S(1)$ is true, i.e., condition (1) is satisfied.

2. Let us assume that $S(k)$ is true for any $k \in N$, that is,

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \quad (A)$$

$$\begin{aligned} S(k+1): 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \frac{(k+1)(k+1+1)(2k+1+1)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \quad (B) \end{aligned}$$

Adding $(k+1)^2$ to both the sides of equation (A), we have

$$\begin{aligned} 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2 + k + 6k + 6)}{6} \\ &= \frac{(k+1)(2k^2 + 7k + 6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \end{aligned}$$

Example 3: Show that $\frac{n^3 + 2n}{3}$ represents an integer $\forall n \in \mathbb{N}$.

Solution: Let $S(n) = \frac{n^3 + 2n}{3}$

1. When $n = 1$, $S(1)$ becomes

$$S(1) = \frac{1^3 + 2(1)}{3} = \frac{3}{3} = 1 \in \mathbb{Z}$$

2. Let us assume that $S(n)$ is true for any $n = k \in \mathbb{W}$, that is,

$$S(k) = \frac{k^3 + 2k}{3} \text{ represents an integer.}$$

Now we want to show that $S(k+1)$ is also an integer. For $n=k+1$, the statement becomes

$$\begin{aligned} S(k+1) &= \frac{(k+1)^3 + 2(k+1)}{3} \\ &= \frac{k^3 + 3k^2 + 3k + 1 + 2k + 2}{3} = \frac{(k^3 + 2k) + (3k^2 + 3k + 3)}{3} \\ &= \frac{(k^3 + 2k) + 3(k^2 + k + 1)}{3} \\ &= \frac{k^3 + 2k}{3} + (k^2 + k + 1) \end{aligned}$$

As $\frac{k^3 + 2k}{3}$ is an integer by assumption and we know that $(k^2 + k + 1)$ is an integer as $k \in \mathbb{W}$

$S(k+1)$ being sum of integers is an integer, thus the condition (2) is satisfied.

Since both the conditions are satisfied, therefore, we conclude by mathematical induction that $\frac{n^3 + 2n}{3}$ represents an integer for all positive integral values of n .

Example 4: Use mathematical induction to prove that

$$3 + 3.5 + 3.5^2 + \dots + 3.5^n = \frac{3(5^{n+1} - 1)}{4} \text{ whenever } n \text{ is non-negative integer.}$$

Solution: Let $S(n)$ be the given statement, that is,

$$S(n): 3 + 3.5 + 3.5^2 + \dots + 3.5^n = \frac{3(5^{n+1} - 1)}{4}$$

The dot (.) between two numbers stands for multiplication symbol.

1. For $n=0$, $S(0)$ becomes $S(0): 3.5^0 = \frac{3(5^{0+1} - 1)}{4}$ or $3 = \frac{3(5-1)}{4} = 3$

Thus $S(0)$ is true i.e., conditions (1) is satisfied.

2. Let us assume that $S(k)$ is true for any $k \in W$, that is,

$$S(k): 3 + 3.5 + 3.5^2 + \dots + 3.5^k = \frac{3(5^{k+1} - 1)}{4} \quad (A)$$

Here $S(k+1)$ becomes

$$\begin{aligned} S(k+1): 3 + 3.5 + 3.5^2 + \dots + 3.5^k + 3.5^{k+1} &= \frac{3(5^{(k+1)+1} - 1)}{4} \\ &= \frac{3(5^{k+2} - 1)}{4} \quad (B) \end{aligned}$$

Adding 3.5^{k+1} on both sides of (A), we get

$$\begin{aligned} 3 + 3.5 + 3.5^2 + \dots + 3.5^k + 3.5^{k+1} &= \frac{3(5^{k+1} - 1)}{4} + 3.5^{k+1} \\ &= \frac{3(5^{k+1} - 1 + 4.5^{k+1})}{4} \\ &= \frac{3[5^{k+1}(1 + 4) - 1]}{4} \\ &= \frac{3(5^{k+2} - 1)}{4} \end{aligned}$$

This shows that $S(k+1)$ is true when $S(k)$ is true. Since both the conditions are satisfied, therefore, by the principle of mathematical induction, $S(n)$ is true for each $n \in W$.

Example 5: Show that $1+3+5+\dots+(2n+5) = (n+3)^2$ for integral values of $n \geq -2$.

Solution:

1. Let $S(n)$ be the given statement, then for $n = -2$, $S(-2)$ becomes,

$$2(-2)+5=(-2+3)^2, \text{ i.e., } 1 = (1)^2 \text{ which is true.}$$

Thus $S(-2)$ is true i.e., the condition (1) is satisfied

2. Let the equation be true for any $n = k \in \mathbb{Z}$, $k \geq -2$, so that

$$1+3+5+\dots+(2k+5) = (k+3)^2 \quad \text{(A)}$$

$$S(k+1): 1+3+5+\dots+(2k+5)+(2k+7) = (k+3)^2 + (2k+7) \quad \text{(B)}$$

Adding $(2k+7)$ on both sides of equation (A) we get,

$$\begin{aligned} 1+3+5+\dots+(2k+5)+(2k+7) &= (k+3)^2 + (2k+7) \\ &= k^2 + 6k + 9 + 2k + 7 \\ &= k^2 + 8k + 16 \\ &= (k+4)^2 \end{aligned}$$

Thus the condition (2) is satisfied. As both the conditions are satisfied, so we conclude that the equation is true for all integers $n \geq -2$.

Example 6: Show that the inequality $4^n > 3^n + 4$ is true, for integral values of $n \geq 2$.

Solution: Let $S(n)$ represents the given statement i.e., $S(n): 4^n > 3^n + 4$ for integral values of $n \geq 2$

1. For $n = 2$, $S(2)$ becomes

$$S(2): 4^2 > 3^2 + 4, \text{ i.e., } 16 > 13 \text{ which is true.}$$

Thus $S(2)$ is true, i.e., the first condition is satisfied.

2. Let the statement be true for any $n = k (\geq 2) \in \mathbb{Z}$, that is

$$4^k > 3^k + 4 \quad (A)$$

Multiplying both sides of inequality (A) by 4, we get

$$\text{or } 4 \cdot 4^k > 4(3^k + 4)$$

$$\text{or } 4^{k+1} > (3+1)3^k + 16$$

$$\text{or } 4^{k+1} > 3^{k+1} + 4 + 3^k + 12$$

$$\text{or } 4^{k+1} > 3^{k+1} + 4 \quad (\because 3^k + 12 > 0) \quad (B)$$

The inequality (B), satisfies the condition (2).

Since both the conditions are satisfied, therefore, by the principle of extended mathematical induction, the given inequality is true for all integers $n \geq 2$.

Use mathematical induction to prove the following formulae for every positive integer n .

1. $1 + 5 + 9 + \dots + (4n - 3) = n(2n - 1)$

2. $1 + 3 + 5 + \dots + (2n - 1) = n^2$

3. $1 + 4 + 7 + \dots + (3n - 2) = \frac{n(3n - 1)}{2}$

4. $1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$

5. $1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{n-1}} = 2 \left[1 - \frac{1}{2^n} \right]$

6. $2 + 4 + 6 + \dots + 2n = n(n + 1)$

7. $2 + 6 + 18 + \dots + 2 \times 3^{n-1} = 3^n - 1$

8. $1 \times 3 + 2 \times 5 + 3 \times 7 + \dots + n \times (2n + 1) = \frac{n(n + 1)(4n + 5)}{6}$

9. $1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + n \times (n + 1) = \frac{n(n + 1)(n + 2)}{3}$

10. $1 \times 2 + 3 \times 4 + 5 \times 6 + \dots + (2n - 1) \times 2n = \frac{n(n + 1)(4n - 1)}{3}$

11. $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n + 1)} = 1 - \frac{1}{n + 1}$

12. $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n - 1)(2n + 1)} = \frac{n}{2n + 1}$

13. $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{(3n - 1)(3n + 2)} = \frac{n}{2(3n + 2)}$

14. $r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}, \quad (r \neq 1)$

15. $a + (a + d) + (a + 2d) + \dots + [a + (n - 1)d] = \frac{n}{2} [2a + (n - 1)d]$

16. $1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + \dots + n \cdot n = \frac{n(n + 1)}{2} - 1$

17. $a_n = a_1 + (n - 1)d$ when, $a_1, a_1 + d, a_1 + 2d, \dots$ form an A.P.

18. $a_n = a_1 r^{n-1}$ when $a_1, a_1 r, a_1 r^2, \dots$ form a G.P.

19. $1^2 + 3^2 + 5^2 + \dots + (2n - 1)^2 = \frac{n(4n^2 - 1)}{3}$

20. $\left(\frac{3}{3} \right) + \left(\frac{4}{3} \right) + \left(\frac{5}{3} \right) + \dots + \left(\frac{n + 2}{3} \right) = \left(\frac{n + 3}{4} \right)$

21. Prove by mathematical induction that for all positive integral values of n

i) $n^2 + n$ is divisible by 2.

ii) $5^n - 2^n$ is divisible by 3.

iii) $5^n - 1$ is divisible by 4.

iv) $8 \times 10^n - 2$ is divisible by 6.

v) $n^3 - n$ is divisible by 6.

22. $\frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n} = \frac{1}{2} \left[1 - \frac{1}{3^n} \right]$

23. $1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1} \cdot n^2 = \frac{(-1)^{n-1} \cdot n(n + 1)}{2}$

24. $1^3 + 3^3 + 5^3 + \dots + (2n - 1)^3 = n^2 [2n^2 - 1]$

25. $x + 1$ is a factor of $x^{2n} - 1; (x \neq -1)$